

Student Record Management System

A command-line Python application to manage student records — add, view, search, update, and delete students. Data is saved permanently in a JSON file so it's not lost when the program closes.

Features

- Add a new student record (Roll Number, Name, Branch, CGPA)
- View all student records
- Search for a student by Roll Number, Name, or Branch
- View students sorted by CGPA (highest first)
- Update an existing student's details
- Delete a student record
- Data persists across runs using a JSON file (`students.json`)
- Input validation: CGPA must be a number between 0 and 10; Roll Number must be numeric
- Handles corrupted/invalid data files gracefully instead of crashing

Technologies Used

- Python 3
- JSON (for file-based data storage)

How It Works

- Each student is stored as a dictionary, with **Roll Number as the key** – this allows fast lookup ($O(1)$ average time) instead of searching through a list one by one.
- All data is loaded from `students.json` when the program starts, and saved back to it after every change, so your data is never lost.

How to Run

1. Make sure Python 3 is installed
2. Download `student.py`
3. Open a terminal in that folder and run:

```
python student.py
```

4. Follow the on-screen menu to add, view, search, update, or delete records

Example Menu

```
1. Add 2. View 3. Search 4. Update 5.  
Delete 6. Exit  
Enter choice:
```

Future Improvements

- Switch from JSON file storage to a proper database (SQLite)
- Add a simple GUI using Tkinter
- Export records to a CSV or PDF report